Test Plan for the Music BoomBoxd



Date	**:**	**17/05/2024**
Version	**:**	**1.0**
State	**:**	**Finished**
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Version History

Version	Date	Author(s)	Changes	State
0.1	10/05/2024	Szymon	Made a beginning on the test plan.	Finished
1.0	17/05/2024	Szymon	Finished the first version of the test plan	Finished
1.1	08/06/2024	Szymon	Updating the TestPlan after the last formative talk with the WKS teacher	Finished

Introduction:

This test plan outlines the testing strategy for the MusicBoomBoxd Windows Forms application. The application allows users to create and edit accounts and also add albums and songs with different user roles with specific functionalities.

Objectives:

The objective of this test plan is to make sure that the application functions correctly and meets the specified requirements in the URS. These tests will cover:

1. Creation and management of Users
2. Creation and management of Songs and Albums
3. Role-specific functionalities
4. Rating system
5. Content display and customization
6. User categorization

The features to be tested:

1. User Management
    a. Creating, viewing, updating, and deleting user profiles (FR1)
    b. Logging in with different user roles (Critic, Musician, Regular user)
    c. User interactions and role-specific functionalities such as:
        - Rating (FR1.3)
        - Adding songs or albums(FR1.4 and FR1.5)
2. Review System
    a. Rating and viewing ratings for songs, albums, and artists (FR2.1, FR2.2)
    b. Filtering content based on ratings, genre, or artist (FR2.2)
3. Content Display and Customization
    a. Displaying recent albums(FR3.1)
    b. Categorizing music by artist, genre, and rating (FR3.2)
    c. Customizing music preferences (FR3.3)
4. User Categorization
    a. Categorizing users as Critics, Regulars, or Musicians (FR4.1)
    b. Prioritizing comments and reviews from Musicians and Critics (FR4.2, FR4.3)

Test environment:

1. Operating System: Windows
2. Development Framework: .NET(C#)
3. Database: SQL Server

Test cases:

User Profile Management (UC1)

TC1.1 User Creation with Valid Data

Objective: Verify that new users can be created with valid data.
Preconditions: The application is running, access to the website.
Test Steps:

1. Navigate to the "Create User" form.
2. Click "Add User".
3. Navigate to the New User Form
4. Enter valid data for all required fields:
    - Enter “ProfilePic.jpg” in the ProfilePicture field.
    - Enter “TestUsername” in the username field.
    - Enter “TestPassword” in the password field.
    - Enter “Test@example.com” in the email field.
    - Enter “Regular” in the usertype field.
    - Enter “20-08-2004” in the birthday field.
    - Enter “Rock” in the favgenre field.
5. Click "Add User".
6. Save the User in the Database

Expected Result (success):

- The system should validate the uniqueness of the username and email.
- The password should be hashed and securely stored.
- The user should be redirected to the login page.

Expected Result (Failure):

- If the username or email already exists, an appropriate error message should be displayed.
- The user is no created and remains on the current form

TC1.2 User Creation with Invalid Data

Objective: Verify that users cannot be created with invalid data.
Preconditions: The application is running.
Test Steps:

1. Navigate to the "Create User" form.
2. Click "Add User".
3. Navigate to the New User Form
4. Enter invalid data (e.g., empty Username, invalid Email format) :
    - Enter "ExistingUser" in the username field (assume this username is already
    - taken).
    - Enter "existing@example.com" in the email field (assume this email is already
    - registered).
5. Click "Save".
6. The system displays an error message
7. Click "Save".
8. System is unable to save the User in the Database

Expected Result (success):

- The user is not created, and an error message is displayed.
- The system displays an error message indicating the username and email are already taken.
- **Expected Result (Failure):**
- The system allows creation with invalid data, which should not happen.

TC1.3 User Login with Valid Credentials

Objective: Verify that users can log in with valid credentials, using their username and password.

Preconditions: At least one user exists in the database.
Test Steps:

1. Navigate to the "Login" form.
2. Enter a valid Username and Password.
    - Enter “TestUsername” in the username field.
    - Enter “TestPassword” in the password field
3. Click "Login".

Expected Result:

- The system should verify the user's credentials.
- The user should be redirected to their profile page and in the Windows forms to the dashboard.
- If the credentials do not match, an appropriate error message should be displayed.

TC1.4 User Login with Invalid Credentials

Objective: Verify that users cannot log in with invalid credentials.

Preconditions: The application is running.

Test Steps:

1. Navigate to the "Login" form.
2. Enter an invalid Username and/or Password.
    - Enter “NonExistingUser” in the username field.
    - Enter “WrongPassword” in the password field
3. Click "Login".
4. The system displays an error message that Username and/or Password is incorrect.

Expected Result: The user is not logged in, and an error message is displayed.

TC1.5 Update User Profile Valid Credentials

Objective: Verify that users can update their profile information.
Preconditions: The user is logged in.
Test Steps:

1. Navigate to the "Profile" form.
2. Edit user details with valid data:
    - Enter “NewUsername” in the username field.
    - Enter “Newemail@gmail.com” in the email field.
3. Click "Add User".

Expected Result: The user's profile information is updated.

TC1.6 Update User Profile Invalid Credentials

Objective: Verify that users cannot update their profile information.
Preconditions: The user is logged in.
Test Steps:

1. Navigate to the "Profile" form.
2. Edit user details with invalid credentials:
    - Enter “” in the username field.
    - Enter “Email” in the email field.
3. Click "Add User".
4. The system displays an error message.

Expected Result: The user's profile information is not updated and an error message is displayed.

TC1.7 Delete User Profile Valid Credentials

Objective: Verify that users can delete their profiles.
Preconditions: The user is logged in.
Test Steps:

1. Navigate to the "User" form in the windows form.
2. Navigate to the “Profile” page on the website.
3. Click "Delete" in the Windows forms.
4. Click "Deactivate Account" on the website.
5. Confirm the deletion.

Expected Result: The user's profile is deactivated, and user is redirected to the login page.

TC1.8 Delete User Profile Invalid Credentials

Objective: Verify that users cannot delete their profiles.
Preconditions: The user is logged in.
Test Steps:

1. Navigate to the "Profile" form.
2. Click "Delete".
3. Confirm the deletion.
4. The system shows an error message.

Expected Result: The user's profile is not deactivated, and an error message is displayed.

Interacting with Reviews (UC2)

Rating (UC2)

TC2.1 Rate Content with Valid Data

Objective: Verify that users can rate songs, albums, and artists..
Preconditions: The user is logged in as a Critic.
Test Steps:

1. Navigate to a song, album, or artist page.
2. Select a song or album or artist.
3. Select a rating from 0 to 5 stars.
4. Click "Submit".

Expected Result: The rating is saved, and the average rating is updated.

TC2.2 Rate Content Invalid

Objective: Verify that users can’t rate songs, albums, and artists..
Preconditions: The user is logged in.
Test Steps:

1. Navigate to a song, album, or artist page.
2. Select a song or album or artist.
3. Select a rating from 0 to 5 stars.
4. Click "Submit".
5. The system shows an error message.

Expected Result: The rating is not saved and an average rating is not updated. The system shows an error message.

TC2.3 View Average Ratings

Objective: Verify that users can view the average ratings of songs, albums, and artists.
Preconditions: The user is logged in.
Test Steps:

1. Navigate to a song, album, or artist page.
2. Select a song or album or artist.
3. View the average rating.

Expected Result: The rating is viewed.

TC2.4 View Average Ratings Invalid

Objective: Verify that users cannot view the average ratings of songs, albums, and artists.
Preconditions: The user is logged in.
Test Steps:

1. Navigate to a song, album, or artist page.
2. Select a song or album or artist.
3. View the average rating.

Expected Result: The rating is not viewed. The system shows an error message.

TC2.5 Filter Content by Ratings

Objective: Verify that users can filter content by ratings, genre, or artist.
Preconditions: The user is logged in.
Test Steps:

1. Navigate to the homepage.
2. Use the filter to find songs, albums, or artists based on ratings, genre, or artist.

Expected Result: The system shows songs, albums, or artists based on the selected filter.

Content and Customization (UC3)

TC3.1 View Recent Albums

Objective: Verify that the system displays the most recent albums and songs.
Preconditions: The user is logged in..
Test Steps:

1. Navigate to the homepage..
2. View the most recent albums and songs.

Expected Result: The most recent albums and songs are displayed..

TC3.2 Search and Categorize Music

Objective: Verify that users can search and categorize music by ratings, genre, or artist.
Preconditions: The user is logged in..
Test Steps:

1. Navigate to the homepage.
2. Use the search function to find music by ratings, genre, or artist.

Expected Result: The system displays the categorized results.

TC3.3 Customize Music Preferences

Objective: Verify that users can customize their music preferences
Preconditions: The user is logged in..
Test Steps:

1. Navigate to the "User" form.
2. Customize music preferences by changing the favorite genre.
3. Enter “Rock” for favorite genre.
4. Click "Save".

Expected Result: The system displays the categorized results.

Admin Login (UC4)

TC4.1 Categorize User

Objective: Verify that users can upgrade their account and be categorized accordingly.
Preconditions: The user is logged in as a Regular user.
Test Steps:

1. Navigate to the "Profile" page.
2. Click "Upgrade Account".
3. Choose the category to upgrade to (Critic or Musician).
4. Click "Confirm".

Expected Result: The user's role is updated based on the selected category.

TC4.2 Musician Adding Content

Objective: Verify that musicians can add songs and albums.
Preconditions: The user is logged in as a Musician.
Test Steps:

1. Navigate to the form to add albums or songs.
2. Fill in the required details.
3. Click "Save".

Expected Result: The song or album is added and displayed on the homepage.

TC4.3 Critic Adding Content

Objective: Verify that critics can add songs and albums.
Preconditions: The user is logged in as a Critic.
Test Steps:

1. Navigate to the form to add albums or songs.
2. Fill in the required details.
3. Click "Save".

Expected Result: The song or album is added and displayed on the homepage.

TC4.4 User Role Functionality

Objective: Verify role-specific functionalities for Critic, Musician, and Regular user.
Preconditions: Users are logged in with respective roles.
Test Steps:

1. Log in as a Critic, Musician, and Regular user separately.
2. Perform role-specific actions:
    - Critic: Add and review content.
    - Musician: Add content.
    - Regular user: View recent albums and songs.

Expected Result: The system allows role-specific actions based on the user role.